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A STEP-BY-STEP MATHEMATICAL DERIVATION OF ANT COLONY OPTIMIZATION ALGORITHM FOR SOLVING 0/1 KNAPSACK PROBLEM

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ABSTRACT

This article provides a meticulous step-by-step mathematical derivation of the Ant Colony Optimization (ACO) algorithm designed for solving the 0/1 Knapsack Problem. Each facet of the algorithm, from initialization to pheromone updates, solution construction, and the delicate balance between exploration and exploitation, is dissected. This comprehensive understanding of ACO's underlying mathematical framework offers a clear roadmap for researchers and practitioners, guiding them in harnessing ACO's power to optimize combinatorial challenges. The 0/1 Knapsack Problem, a significant challenge in computer science due to its NP-hard nature, involves strategically loading objects into a knapsack with limited capacity. Despite its computational complexity, this problem finds applications in operations research and finance. The article addresses the urgency for swift solutions by exploring heuristic algorithms inspired by ant behavior, providing a valuable guide for efficiently solving the 0/1 Knapsack Problem.

Keywords: Ant Colony Optimization, 0/1 Knapsack Problem, Combinatorial Optimization, Mathematical Derivation, Heuristic Algorithms.

I. INTRODUCTION

A. Background

The 0/1 Knapsack Problem stands as a classic example of a combinatorial optimization challenge, where items with varying weights and values must be selected to maximize the overall value while adhering to a predefined weight constraint. Ant Colony Optimization, inspired by the foraging behavior of ants, presents a robust approach to tackle such complex problems [1, 2, and 3]. The 0/1 knapsack problem is a widely used problem due to its practical importance [4]. The 0/1 Knapsack problem consists of loading objects in to a knapsack load capacity. Each object can be loaded or not loaded into the knapsack, relating 0-1 decision about object loading [5]. The 0/1 knapsack problem does not allow the user to put multiple copies of the same items in their knapsack. The 0/1 knapsack problem is also well known in computer science, as it belongs to a class of problems which are NP-hard. When a problem is “NP-hard” there is no algorithm to solve the problem in polynomial time. It means that if there were a polynomial time solution of other NP problems could be reduced to the 0/1 knapsack problem in polynomial time. The 0/1 knapsack problem has applications are in different areas such as operations research and finance. In order to obtain the solution for the 0/1 knapsack problem a little period of time, heuristic algorithms are used [14, 15]. Such algorithms, which are based on the behavior of ants, are taken into consideration in this article.

B. Objective

This manuscript aims to demystify the mathematical derivation of the Ant Colony Optimization algorithm for the specific application of solving the 0/1 Knapsack Problem. By breaking down each step, we provide a clear and concise guide for researchers and practitioners to understand the underlying mathematical intricacies.

II. METHODOLOGY

A. Mathematical formalization

The mathematical model of the 0-1 knapsack problem can be illustrated as:

Maximize $\sum_{i=1}^{n} Z_i X_i$

Subjected to constraints: $\sum_{i=1}^{n} W_i X_i \leq V_c$

A knapsack will have a capacity ' V_c',there will be N distinct object (i) with weight(W_i) and profits(Z_i) are to be placed into the knapsack such that the total profit is maximum and the total weights of the objects should be

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lesser than or equal to the capacity of the knapsack,.X_i takes the values 0/1, 1 specifies that the object' i 'are considered into the knapsack and 0 specifies that the object 'i' is not considered into the knapsack. Each object X_i has its own profit Z_i and its own weight W_i . The set N={$x_1, x_2,\ldots.. x_n$} is the set of available objects which can be loaded into the knapsack [16, 17]. When an object x_i goes into the knapsack, then capacity is decreased by the weight of this loaded object x_i . This new value for the available knapsack capacity is called the current knapsack capacity V_c . If the weight of any object is higher than the current knapsack capacity V_c , then that object cannot be loaded into the knapsack and is deleted from the list N. After an object x_i is loaded into the knapsack, a new list N_j has to be compiled. This new list N_j is obtained from the preceding list N_i. The solution to the problem constitutes the objects loaded into the knapsack, that object which has been added in the set [4-13].

B. Greedy Algorithm

Greedy is an algorithm design strategy which makes an optimal solution in stages. At each stage we make a decision that appears to be best at the time. A decision made in one stage is not changed in a later stage. The heuristic search for solving the 0/1 knapsack problem is find the sorted order of objects with respect to profit weight ratio. Then consider the object in that order by considering the constraint i.e., sum of the weights of the considered objects should be lesser than or equal to the capacity of the knapsack.

There are four steps to solve greedy algorithm 0/1 knapsack problem

Step-1 From the remaining objects, select the object with maximum profit that fits into the knapsack, the object with the largest profit is packed first (provided enough capacity is available), then the one with next largest profit, and so on.

Step-2 From the remaining objects, select the one that has minimum weight and also fits into the knapsack.

Step-3 From the remaining objects, Select the one with maximum $\frac{Z_i}{W_i}$ that fits into the knapsack, Place it in the knapsack else Proceed to the next one.

Step-4 calculates the sum of profit, each selecting objects.

Example: consider the knapsack instance n=7,W_i={2,2,5,7,1,4,1}, P_i={10,5,15,7,6,18,3},V_C=14

Table 1: Greedy algorithm 0/1 Knapsack problem Computation of results

Object	1	2	3	4	5	6	7
Profit	10	5	15	7	6	18	3
Weight(N_i)	2	2	5	7	1	4	1
Z_i/W_i	5	2.5	3	1	6	4.5	3
x_i	1	1	1	0	1	1	0
Set N_j	x_1	x_2	x_3	x_4	x_5	x_6	x_7

The solution x_i ={1,1,1,0,1,1,0}. The total profit from this solution is= 6+10+18+15+5=54.

III. MODELING AND ANALYSIS

A. Ant System for 0/1 knapsack problem

The problem is represented by graph and the ants go on the graph to construct solutions. The solutions are represented by edges in the graph G(V,E) where V is the set of vertices(objects) and E is the set of edges[18-20]. After the initialization of pheromone trail:

$$\tau_{ij} = \begin{cases} 1, & (i,j) \in N_i \\ 0, & (i,j) \notin N_i \end{cases}$$

,the ants construct feasible solutions, starting from random nodes, at each step the ants moves and select the best one (according to some probabilistic rules)to continue the rest of the tour by maximum P_{ij} to choose the node j when the current node is i, is based on the heuristic information(attractive of ant towards the solution) η_{ij}[1-3]:

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$$\eta_{ij} = \frac{\frac{Z_i}{W_i}}{V_C} \quad (1)$$

Where V_c capacity of knapsack. Where i, j =1,2,……., n.

At the beginning of the ants work the knapsack is empty; thus the set N_j is also empty (N_j=∅). Next, all objects from the set N are verified in terms of their weights and the knapsack load capacity. A new object x_i can be selected to be loaded into the knapsack only from those objects whose weight is less than the current knapsack load capacity. After a selected object x_i has been packed into the knapsack, the current knapsack capacity is reduced, then the object x_i inside the knapsack $N_j = \{x_i\}$. After 'j' steps, other objects are loaded into the knapsack, so the contents of the set

$$N_j = \{x_1, x_2, \ldots\ldots, x_n\}. \quad (2)$$

For all ants have completed their turn, the pheromone evaporation is implemented as follows:

$$\boldsymbol{\tau_{ij} = (1-\rho)\,\tau_{ij}\ ,\ j \in N_i} \quad (3)$$

Where $\rho \in (0,1)$ is a parameter which simulates the evaporation *rate* of the pheromone intensity.

After all the ants have completed their turns, the pheromone trails are updated as follows:

$$\tau_{ij} = \tau_{ij} + \sum_{k=1}^{n} \Delta\tau_{ij}^{k} \quad , \ j \in N_i \quad (4)$$

Where $\Delta\tau_{ij}^{k}$ is the amount of pheromone deposited by ant k on visited arcs.

It is defined as follows:

$$\Delta\boldsymbol{\tau_{ij}^{k}} = \frac{1}{1+Z_{best}-Z_i} \quad (5)$$

Where Z_{best} is the maximum profit of given objects and Z_i is the profit of the object of current state, $N_i is$ the feasible neighborhood of state i.

B. ACO algorithm for 0/1 Knapsack Problem

Step-1 Initialize pheromone τ for all objects and parameter α,β also define.

Step-2 Distribute weight, profit and capacity to all nodes.

Step-3 Calculate the heuristic information for ants move η_{ij} at every node for all objects respect to Capacity (as equation-1).

Step-4Calculate the probability $P_{ij} = \frac{\tau_{ij}^{\alpha}\,\eta_{ij}^{\beta}}{\sum_{j\in N_i^k}\tau_{ij}^{\alpha}\,\eta_{ij}^{\beta}}$,at every node for all objects. (Where N_i^k is the feasible neighborhood of ant k)

Step-5 At every node find maximum P_{ij}

Step-6 If the knapsack capacity V_c is higher than objects weight w_i then put object in knapsack and reduce the capacity of knapsack by weight ($V_c = V_c - w_i$) otherwise go to step-12.

Step-7 Find the weight, profit corresponding maximum P_{ij} and update the status.

Step-8 For each selected object calculate the profit ($Z_{best} = Z_{best} + Z_i$) and send it to master node.

Step-9 Update pheromone evaporation of each object at every node (as equation-3).

Step-10 Update the pheromone of selected object at every node respect to capacity.(as equation-4&5).

Step-11 Repeat the step-2 to step-11 until all knapsack capacity not fills.

Step-12 Calculate the overall profit at master node.

IV. RESULTS AND DISCUSSION

A. Numerical Evaluation of ACO algorithm for 0/1 Knapsack Problem

Example: consider the knapsack instance n=7,W_i={2,2,5,7,1,4,1}, P_i={10,5,15,7,6,18,3},V_C=14

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B. Initialization of parameters:

Assume that $\alpha = \beta = 1$ and $\rho = 0.5$,k =7 (i.e. number of ants),Z=0 and neighborhood size is the entire graph.

Initialization of pheromone:

$$\tau_{ij} = \begin{cases} 1, & (i,j) \in N_i \\ 0, & (i,j) \notin N_i \end{cases}$$

C. A Step by Step Illustration

Iteration 1:

Initially, put a single ant at each object.

Table 2: ACO-0/1 Knapsack problem Computation results iteration-1

Object	1	2	3	4	5	6	7
Profit(Z_i)	10	5	15	7	6	18	3
Weight(w_i)	2	2	5	7	1	4	1
Pheromone(τ)	1	1	1	1	1	1	1
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	0.35	0.17	0.21	0.07	0.42	0.32	0.21

Then calculate probability

$$P_{11}^1 = \frac{\tau_{11}\eta_{11}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.35}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.35}{1.75} = 0.2$$

$$P_{12}^1 = \frac{\tau_{12}\eta_{12}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.17}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.17}{1.75} = 0.09$$

$$P_{13}^1 = \frac{\tau_{13}\eta_{13}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.21}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.21}{1.75} = 0.12$$

$$P_{14}^1 = \frac{\tau_{14}\eta_{14}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.07}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.07}{1.75} = 0.04$$

$$P_{15}^1 = \frac{\tau_{15}\eta_{15}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.42}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.42}{1.75} = 0.24$$

$$P_{16}^1 = \frac{\tau_{16}\eta_{16}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.32}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.32}{1.75} = 0.18$$

$$P_{17}^1 = \frac{\tau_{17}\eta_{17}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}+\tau_{17}\eta_{17}} = \frac{0.21}{0.35+0.17+0.21+0.07+0.42+0.32+0.21} = \frac{0.21}{1.75} = 0.12$$

The highest probability is on the object x_5 is 0.24

Hence select the object whose weight is 1 and its profit is 6.

$$V_c = V_c - w_i = 14-1=13$$

$$\text{Profit (Z)} = 0+6=6$$

Then update of Pheromone Trail:

Calculate the pheromone evaporation rate using formula: $\boldsymbol{\tau_{ij} = (1-\rho)\,\tau_{ij}}$

so $\tau_{11} = 0.5$, $\tau_{12} = 0.5$, $\tau_{13} = 0.5$, $\tau_{14} = 0.5$, $\tau_{15} = 0.5$, $\tau_{16} = 0.5$,here $Z_{best} = 18$.

$$\Delta\tau_{11} = \frac{1}{1+18-10} = \frac{1}{9} = 0.11, \quad \tau_{1\,1} = 0.5 + 0.11 = 0.61$$

$$\Delta\tau_{12} = \frac{1}{1+18-5} = \frac{1}{14} = 0.07, \quad \tau_{1\,2} = 0.5 + 0.07 = 0.57$$

$$\Delta\tau_{13} = \frac{1}{1+18-15} = \frac{1}{4} = 0.25, \quad \tau_{1\,3} = 0.5 + 0.25 = 0.75$$

$$\Delta\tau_{14} = \frac{1}{1+18-7} = \frac{1}{12} = 0.08, \quad \tau_{1\,4} = 0.5 + 0.08 = 0.58$$

$$\Delta\tau_{15} = \frac{1}{1+18-18} = \frac{1}{1} = 1, \quad \tau_{1\,5} = 0.5 + 1 = 1.5$$

$$\Delta\tau_{16} = \frac{1}{1+18-3} = \frac{1}{10} = 0.06, \quad \tau_{1\,6} = 0.5 + 0.25 = 0.75$$

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Iteration 2 ($V_c = 13$)

Table 3: ACO-0/1 Knapsack problem Computation results iteration-2

Object	1	2	3	4	5	6
Profit(Z_i)	10	5	15	7	18	3
Weight(w_i)	2	2	5	7	4	1
Pheromone(τ)	0.61	0.57	0.75	0.58	1.5	0.75
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	0.38	0.19	0.23	0.07	0.34	0.23

Then calculate probability

$$P_{11}^2 = \frac{\tau_{11}\eta_{11}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}}$$

$$= \frac{(0.61).(0.38)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.2318}{1.2357} = 0.18$$

$$P_{12}^2 = \frac{\tau_{12}\eta_{12}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}}$$

$$= \frac{(0.57).(0.19)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.1083}{1.2357} = 0.08$$

$$P_{13}^2 = \frac{\tau_{13}\eta_{13}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}}$$

$$= \frac{(0.75).(0.23)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.1725}{1.2357} = 0.13$$

$$P_{14}^2 = \frac{\tau_{14}\eta_{14}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}} =$$

$$\frac{(0.58).(0.07)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.0406}{1.2357} = 0.03$$

$$P_{15}^2 = \frac{\tau_{15}\eta_{15}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}} =$$

$$\frac{(1.5).(0.34)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.51}{1.2357} = 0.41$$

$$P_{16}^2 = \frac{\tau_{16}\eta_{16}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}+\tau_{16}\eta_{16}}$$

$$= \frac{(0.75).(0.23)}{(0.61).(0.38)+(0.57).(0.19)+(0.75).(0.23)+(0.58).(0.07)+(1.5).(0.34)+(0.75).(0.23)} = \frac{0.1725}{1.2357} = 0.13$$

The highest probability is 0.41

Hence select the object whose weight is 4 and its profit is 18.

$V_c = V_c - w_i = 13-4=9$

Profit (Z) = 6+18=24

Then update of Pheromone Trail:

Calculate the pheromone evaporation rate using formula: $\boldsymbol{\tau_{ij} = (1-\rho)\,\tau_{ij}}$

so τ_{11}= (0.5).(0.61)= 0.30 , τ_{12} =(0.5).(0.57)=0.28, τ_{13} =(0.5).(0.75)=0.37,

τ_{14} =(0.5).(0.58)=0.29, τ_{15} =(0.5).(0.75)=0.37 , Z_{best} = 15.

$$\Delta\tau_{11} = \frac{1}{1+15-10} = \frac{1}{6} = 0.16,\ \tau_{1\,1} = 0.30 + 0.16 = 0.46$$

$$\Delta\tau_{12} = \frac{1}{1+15-5} = \frac{1}{10} = 0.1,\ \tau_{1\,2} = 0.28 + 0.1 = 0.38$$

$$\Delta\tau_{13} = \frac{1}{1+15-15} = \frac{1}{1} = 1,\ \tau_{1\,3} = 0.37 + 1 = 1.37$$

$$\Delta\tau_{14} = \frac{1}{1+15-7} = \frac{1}{9} = 0.11,\ \tau_{1\,4} = 0.29 + 0.11 = 0.40$$

$$\Delta\tau_{15} = \frac{1}{1+15-3} = \frac{1}{13} = 0.07,\ \tau_{1\,5} = 0.37 + 0.07 = 0.44$$

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Iteration 3 ($V_c = 9$)

Table 4: ACO-0/1 Knapsack problem Computation results iteration-3

Object	1	2	3	4	5
Profit(Z_i)	10	5	15	7	3
Weight(w_i)	2	2	5	7	1
Pheromone(τ)	0.46	0.38	1.37	0.40	0.44
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	0.55	0.27	0.33	0.11	0.33

Then calculate probability

$$P_{11}^3 = \frac{\tau_{11}\eta_{11}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}} = \frac{(0.46).(0.55)}{(0.46).(0.55)+(0.38).(0.27)+(1.37).(0.33)+(0.40).(0.11)+(0.44).(0.33)} = \frac{0.233}{0.9969} = 0.25$$

$$P_{12}^3 = \frac{\tau_{12}\eta_{12}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}} = \frac{(0.38).(0.27)}{(0.46).(0.55)+(0.38).(0.27)+(1.37).(0.33)+(0.40).(0.11)+(0.44).(0.33)} = \frac{0.1026}{0.9969} = 0.10$$

$$P_{13}^3 = \frac{\tau_{13}\eta_{13}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}} = \frac{(1.37).(0.33)}{(0.46).(0.55)+(0.38).(0.27)+(1.37).(0.33)+(0.40).(0.11)+(0.44).(0.33)} = \frac{0.4521}{0.9969} = 0.45$$

$$P_{14}^3 = \frac{\tau_{14}\eta_{14}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}} = \frac{(0.40).(0.11)}{(0.46).(0.55)+(0.38).(0.27)+(1.37).(0.33)+(0.40).(0.11)+(0.44).(0.33)} = \frac{0.044}{0.9969} = 0.04$$

$$P_{15}^3 = \frac{\tau_{15}\eta_{15}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}+\tau_{15}\eta_{15}} = \frac{(0.44).(0.33)}{(0.46).(0.55)+(0.38).(0.27)+(1.37).(0.33)+(0.40).(0.11)+(0.44).(0.33)} = \frac{0.1452}{0.9969} = 0.14$$

The highest probability is 0.45

Hence select the object whose weight is 5 and its profit is 15.

$$V_c = V_c - w_i = 9-5=4$$

$$\text{Profit (Z)} = 24+15=39$$

Then update of Pheromone Trail:

Calculate the pheromone evaporation rate using formula: $\boldsymbol{\tau_{ij} = (1-\rho)\,\tau_{ij}}$

so $\tau_{11} = (0.5).(0.46) = 0.23$, $\tau_{12} = (0.5).(0.38) = 0.19$, $\tau_{13} = (0.5).(0.40) = 0.2$,

$\tau_{14} = (0.5).(0.44) = 0.22$, $Z_{best} = 10$.

$$\Delta\tau_{11} = \frac{1}{1+10-10} = \frac{1}{1} = 1, \quad \tau_{1\,1} = 0.23 + 1 = 1.23$$

$$\Delta\tau_{12} = \frac{1}{1+10-5} = \frac{1}{6} = 0.16, \quad \tau_{1\,2} = 0.19 + 0.16 = 0.35$$

$$\Delta\tau_{13} = \frac{1}{1+10-7} = \frac{1}{4} = 0.25, \quad \tau_{1\,3} = 0.2 + 0.25 = 0.45$$

$$\Delta\tau_{14} = \frac{1}{1+10-3} = \frac{1}{8} = 0.12, \quad \tau_{1\,4} = 0.22 + 0.12 = 0.34$$

Iteration 4 ($V_c = 4$)

Table 5: ACO-0/1 Knapsack problem Computation results iteration-4

Object	1	2	3	4
Profit(Z_i)	10	5	7	3
Weight(w_i)	2	2	7	1
Pheromone(τ)	1.23	0.35	0.45	0.34
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	1.25	0.625	0.25	0.75

Then calculate probability

$$P_{11}^4 = \frac{\tau_{11}\eta_{11}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}} = \frac{(1.23).(1.25)}{(1.23).(1.25)+(0.35).(0.625)+(0.45).(0.25)+(0.34).(0.75)} = \frac{1.5375}{2.12375} = 0.72$$

$$P_{12}^4 = \frac{\tau_{12}\eta_{12}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}} = \frac{(0.35).(0.625)}{(1.23).(1.25)+(0.35).(0.625)+(0.45).(0.25)+(0.34).(0.75)} = \frac{0.21875}{2.12375} = 0.10$$

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$$P_{13}^4=\frac{\tau_{13}\eta_{13}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}}=\frac{(0.45).(0.25)}{(1.23).(1.25)+(0.35).(0.625)+(0.45).(0.25)+(0.34).(0.75)}=\frac{0.1125}{2.12375}=0.05$$

$$P_{14}^4=\frac{\tau_{14}\eta_{14}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}+\tau_{14}\eta_{14}}=\frac{(0.34).(0.75)}{(1.23).(1.25)+(0.35).(0.625)+(0.45).(0.25)+(0.34).(0.75)}=\frac{0.255}{2.12375}=0.12$$

The highest probability is 0.72

Hence select the object whose weight is 2 and its profit is 10.

$$V_c = V_c - w_i = 4-2=2$$

$$\text{Profit (Z)} = 39+10=49$$

Then update of Pheromone Trail:

Calculate the pheromone evaporation rate using formula: $\boldsymbol{\tau_{ij} = (}1-\rho)\,\boldsymbol{\tau_{ij}}$

so $\tau_{11} = (0.5).(0.35)= 0.175$, $\tau_{12} =(0.5).(0.45)=0.225$, $\tau_{13} =(0.5).(0.34)=0.17$, $Z_{best} = 7.$

$$\Delta\tau_{11} = \frac{1}{1+7-5}=\frac{1}{3}=0.33,\ \tau_{1\,1} = 0.175 + 0.33=0.50$$

$$\Delta\tau_{12} = \frac{1}{1+7-7}=\frac{1}{1}=1,\ \tau_{1\,2} = 0.225 + 1=1.22$$

$$\Delta\tau_{13} = \frac{1}{1+7-1}=\frac{1}{7}=0.142,\ \tau_{1\,3} = 0.17 + 0.14=0.31$$

Iteration 5 ($V_c = 2$)

Table 6: ACO-0/1 Knapsack problem Computation results iteration-5

Object	1	2	3
Profit(Z_i)	5	7	3
Weight(w_i)	2	7	1
Pheromone(τ)	0.50	1.22	0.31
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	1.25	0.5	1.5

Then calculate probability

$$P_{11}^5=\frac{\tau_{11}\eta_{11}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}}=\frac{(0.50).(1.25)}{(0.50).(1.25)+(1.22).(0.5)+(0.31).(1.5)}=\frac{0.625}{1.7}=0.36$$

$$P_{12}^5=\frac{\tau_{12}\eta_{12}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}}=\frac{(1.22).(0.5)}{(0.50).(1.25)+(1.22).(0.5)+(0.31).(1.5)}=\frac{0.61}{1.7}=0.35$$

$$P_{13}^5=\frac{\tau_{13}\eta_{13}}{\tau_{11}\eta_{11}+\tau_{12}\eta_{12}+\tau_{13}\eta_{13}}=\frac{(0.31).(1.5)}{(0.50).(1.25)+(1.22).(0.5)+(0.31).(1.5)}=\frac{0.465}{1.7}=0.27$$

The highest probability is 0.36

Hence select the object whose weight is 2 and its profit is 5.

$$V_c = V_c - w_i = 2-2=0$$

$$\text{Profit (Z)} = 49+5=54$$

Then update of Pheromone Trail:

Calculate the pheromone evaporation rate using formula: $\boldsymbol{\tau_{ij} = (}1-\rho)\,\boldsymbol{\tau_{ij}}$

so $\tau_{11} = (0.5).(1.22)= 0.61$, $\tau_{12} =(0.5).(0.31)=0.15$, $Z_{best} = 7.$

$$\Delta\tau_{11} = \frac{1}{1+7-7}=\frac{1}{1}=1,\ \tau_{1\,1} = 0.61 + 1=1.61$$

$$\Delta\tau_{12} = \frac{1}{1+7-3}=\frac{1}{5}=0.2,\ \tau_{1\,2} = 0.15 + 0.2=0.35$$

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Iteration 6 ($V_c = 0$)

Table 7: ACO-0/1 Knapsack problem Computation results iteration-6

Object	1	2
Profit(Z_i)	7	3
Weight(w_i)	7	1
Pheromone(τ)	1.61	0.35
$\eta_{ij} = \frac{Z_i}{\frac{w_i}{v_c}}$	Undefined	undefined

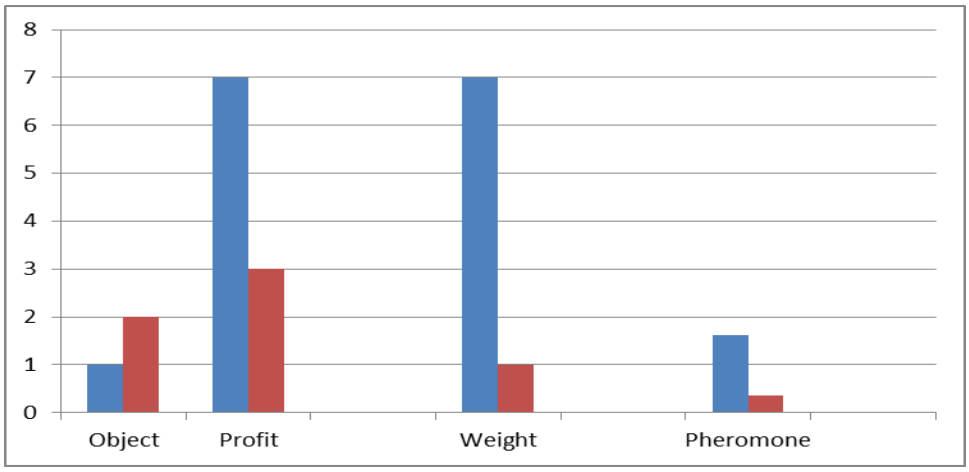


Figure1: ACO-0/1 Knapsack problem Computation results

Profit (Z) = 54, as V_c =0, stop the execution of algorithm.

The total profit from this solution is 54. This research solves 0/1 knapsack problem with single objective, based on Ant Colony Algorithm to solve the multi-dimensional 0/1 Knapsack problem. In the end, Figure 1 is depicts illustrate of the process of the algorithm and illustrated the algorithm how it is feasible and easy to understand.

V. CONCLUSION

In conclusion, this article successfully applies the Ant Colony Algorithm to address the single-objective 0/1 Knapsack Problem and extends its utility to the multi-dimensional variant. The meticulous mathematical derivation of the Ant Colony Optimization (ACO) algorithm provides a robust foundation for understanding and implementing this efficient optimization technique. Illustrated through a practical example, the algorithm proves both feasible and comprehensible. Future work should focus on optimizing performance metrics, exploring adaptability to dynamic problem instances, and incorporating parallel or hybrid approaches for enhanced scalability. Ongoing empirical validation and comparisons with alternative optimization techniques will further elucidate ACO's potential, contributing to its advancement in tackling complex combinatorial optimization challenges.

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